

Siwon Lee

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EDUCATION

Washington University in St. Louis

Ph.D. in Economics

MO, USA

Aug 2024 – Present

- Research Fields: Financial Markets, Asset Pricing, Exchange Rate, International Trade

Washington University in St. Louis

M.A. in Economics

MO, USA

Dec 2025

Sungkyunkwan University

B.A. in Global Economics, B.S. Mathematics

Seoul, Korea

Mar 2021 – Aug 2024

- Math: Real Analysis, Basic Mathematics for AI, Differential Geometry, Probability Theory, Computer Applied Mathematics
- Software: Operating System, Intro to DB, Data Structures, Intro to AI, System Programming, Algorithm
- Economics: Adv Macroeconomic Theory, Game Theory, Econometrics, Mathematical Economics, Intro Financial Accounting

LUSEM & HANKEN University

Winter Joint Exchange Program

Sweden and Finland

Jan 2023

- Visited KPMG corporation in Helsinki and discussed sustainability
- Discussed critical thinking and examples of sustainability in each country

TEACHING EXPERIENCE

Assistant Instructor

WashU, USA

Aug 2025 – Present

- Introduction to Macroeconomics
- Introduction to Microeconomics

Teaching Assistant

SKKU, Korea

Feb 2023 – Jun 2024

- Algorithm and Problem Solving
- Computer Thinking and SW coding
- Problem Solving

Young Economic Education Supporters

Lecturer

Bank of Korea, Korea

Apr 2023 – Dec 2023

Sungkyun Tutoring

SKKU, Korea

Feb 2022 – Jun 2024

- Intermediate Microeconomics Theory
- Intermediate Macroeconomics Theory
- Econometrics

REVIEW EXPERIENCE

20th Economics Graduate Student Conference

Reviewer

WashU, USA

Aug 2025

21st Economics Graduate Student Conference

Reviewer

WashU, USA

Aug 2026

PRESENTATION

The Effect of Internet Reviews on the Demand of Experience Goods: Case of Korean Movie Industry

Sungkyunkwan-KEIO Symposium

Jan 2024

Stabilizing ETF Price Gaps through Liquidity Provision with AI

2022 Capstone Design and AI Hackathon

Oct 2022

Forecasting Inflation and Economic Recession: Comparison of Econometrics and Machine Learning

2022 SKKU Annual Conference on Economics

Aug 2022

WORK IN PROGRESS

From Tariffs to Portfolios: Trade Policy Uncertainty and Household Asset Allocation

Author

Second Year Paper

- Provided empirical evidence linking tariff policy uncertainty to household portfolio allocation using local projections
- Developed a small open economy GE model with CTMC to analyze transmission of tariff shocks to asset allocation
- Incorporated a Merton portfolio framework with inventory-based production structure and fiscal policy interactions
- Quantified how tariff policy uncertainty affects expected returns and portfolio risk across major stock, bond, and cash

Domestic Firm Financial Vulnerability Consequences of Exchange Rate Depreciation

Author

Dissertation Proposal

- Established empirical relationships between exchange rate depreciation and firm-level financial vulnerability using LP
- Identified heterogeneous effects driven by firm characteristics, including trade openness and financial hedging behavior
- Implemented Deep Equilibrium Networks to solve equilibrium conditions in settings where convexity is not guaranteed
- Developed a structural model with endogenous exchange rates and firm heterogeneity to capture vulnerability dynamics

Asset Allocation under Tariff Policy Uncertainty: A Fourier-Embedded DeepONet Approach with SDE

Author

Draft

- Developed a stochastic differential equation framework with jump diffusion to model asset dynamics under TPU shocks
- Characterized the investor's optimal portfolio problem using Itô's Lemma under stochastic return and volatility processes
- Implemented a Fourier-Embedded DeepONet under a CTMC regime to approximate high-dimensional policy functions
- Demonstrated substantial computational efficiency gains relative to traditional numerical calibration approaches

Market Impact of Large Shareholder Stock Sales: An Agent-Based Approach

Author

Draft

- Simulated liquidity shocks and volatility responses arising from large shareholder stock sales in financial markets
- Implemented agent-based models to capture heterogeneous investor behavior in limit order-book trading environments
- Analyzed how information asymmetry and heterogeneous risk preferences amplify price impact and volatility dynamics

Labor Market Conditions and Household Financial Vulnerability

Author

Research Proposal

- Constructed a state-year panel combining labor market indicators with mortgage delinquency data across regions
- Identified predictive relationships between labor market vulnerability measures and household financial distress outcomes
- Provided evidence supporting a causal link using lagged specifications and robustness checks across model variations

COMPLETED RESEARCH PROJECTS

Analyzing the Impact of Monetary Policy on Construction and Facility Investment

Co-author

SKKU, Korea

Mar 2024 – Jul 2024

Analyzing the Effect of Monetary Policy on Technological Progress: NAICS 334

Team Leader

DB, Korea

Dec 2023 – Feb 2024

Comparison of South Korea's 1998 and 2008 Economic Crises

Team Member

SKKU, Korea

Feb 2023 – Jun 2023

Numerical Investigation for Economic Modeling Using Python

Team Member

SKKU, Korea

Jun 2022 – Jul 2022

Analyzing the Relationship Between COVID-19 and Trade

Team Leader

SKKU, Korea

Apr 2022 – Jun 2022

Forecasting Changing Phase of Financial Market When Implementation CBDC

Co-author

Korea

Feb 2022

Implementation of CBDC to Reduce Public Antipathy Toward Privacy Violations

Co-author

SKKU, Korea

Aug 2021

Project of Convergence Basics

Team Member

SKKU, Korea

Jun 2021 – Aug 2021

Calculated Production Possibility Frontier

Team Member

SKKU, Korea

Apr 2021 – Jun 2021

WORK EXPERIENCE

TRC Research Assistant

Research Assistant

Wisdomagora

Editor Management Intern

UHK, Remote

Nov 2023 – Jan 2024

Seoul, Korea

Sep 2022 – Feb 2023

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EXTRACURRICULAR ACTIVITIES

Transdisciplinary Mentoring Community

Mentee

WashU, USA

Aug 2024 – Jul 2025

Global Leaders in Finance(GLIF)

Team Member

Seoul, Korea

Mar 2021 – Mar 2022

MIMIC

Team Member

SKKU, Korea

Sep 2022 – Jun 2024

Seekers

President

SKKU, Korea

Mar 2022 – Jun 2024

AWARDS

2022 SKKU Annual Conference on Economics

Top Prize

SKKU, Korea

Aug 2022

2022 Capstone Design and AI Hackathon

Encouragement Prize

The Korean Association of Computer Education

Oct 2022

Learning-Fair

College Dept. Dean's Awards

SKKU, Korea

Nov 2021

Sungkyun Tutoring

Excellence Prize

SKKU, Korea

Dec 2022

Facebook Hacker Cup 2021

Round 2

Facebook, Online

Sep 2021

Big Data Analysis Contest by NH Investment

Participation Prize

NH Investment

Nov 2022

Dean's List

SKKU

Aug 2022

SCHOLARSHIP

Fellowship & Full Scholarship

Aug 2024 – Aug 2029

WashU

Scholarship for Extracurricular Challenge

Jan 2024

SKKU

Scholarship for Global Learning Leader

Dec 2023

SKKU

Scholarship for Global Exchange Student

Mar 2023

SKKU

Scholarship for Academic Excellence

Aug 2022

SKKU

Scholarship for Academic Excellence

Feb 2022

SKKU

State Scholarship

Feb 2021

Government of South Korea

ADDITIONAL INFORMATION

Achievements: ADsP, Certification of Cody Joint Event(MATLAB)

Skills: Programming(C, Python, MYSQL, MATLAB), LaTeX, MS Office(PowerPoint, Excel)

Languages: Korean(Native), English(Fluent)

Interests: International economics, Finance, Exchange Rate, Machine Learning

Hobbies: Table Tennis, Tennis, Listening to Music, Basketball